

POC - Automated PDF Document Extraction & Summarization

- 1. Introduction
- 2. Objectives & Scope
- 3. Business Case
- 4. Extraction Attributes
- 5. Key Components
- 6. Pipeline Workflow
- 7. Engineering Details
 - 7.1 PDF Text & Image Extraction (pdf_extractor.py)
 - 7.2 LLaMA API Wrapper (call_llama_api.py)
 - 7.3 Orchestration & Prompting (pdf_pipeline.py)
 - 7.4 Interactive UI (gradio_app.py)
- 8. Results & Evaluation
- 9. Roadmap / ToDo
- 10. Internal Commands to turn off/on corn Job:

1. Introduction

This document describes the proofofconcept (POC) pipeline that automatically extracts text and images from PDF documents, performs OCR where needed, and uses an LLM (LLaMA3.18BInstruct) to generate concise summaries and document classifications.

2. Objectives & Scope

Objectives

- Extract both textual and visual content from PDFs, including scanned pages.
- Generate short summaries (23 sentences) and assign each document to a single predefined category via LLM.
- Provide an interactive web interface (Gradio) to upload PDFs and inspect results for demo purposes.

InScope

- Text extraction via [Docling](#) (with OCR fallback).
- Image extraction and captioning via [PikePDF](#) + BLIP.
- LLMdriven summarization and classification.
- Lowlatency, resourceaware execution suitable for demo environments.

OutofScope / Future Work

- Support for additional file types (DOC/DOCX, images, etc.).
- Expanded category taxonomy.

3. Business Case

The pipeline enables a “scantocuration” workflow: a user photographs a document, converts it to PDF, and immediately receives searchable text, image captions, summary, and classification. This accelerates intelligent indexing of personal documents and prepares content for downstream search and analytics.

4. Extraction Attributes

For every PDF we capture:

- Ordered text blocks / paragraphs
- Extracted images with BLIP captions (conditional; see §7.1)
- OCR text for scanned pages
- Metadata: filename, page count, word count, extraction timestamp, element type, page number

5. Key Components

Open Source Libraries	Role
Docling , GitHub	Extract digital text & lightweight OCR
PikePDF , GitHub	Lossless image extraction
BLIP	Caption generation for images

Pipeline Components	Role
<code>pdf_extractor.py</code>	Wraps Docling + PikePDF/BLIP and emits a unified JSON
<code>call_llama_api.py</code>	Thin wrapper around internal LLaMA endpoint
<code>pdf_pipeline.py</code>	Orchestrates extraction prompt creation LLM call artifact storage
<code>gradio_app.py</code>	Browserbased UI for batch uploads & result exploration

6. Pipeline Workflow

```
User → Gradio UI
      → pdf_pipeline.py
         → pdf_extractor.py
            → Docling (text + OCR)
            → PikePDF + BLIP (images + captions)
            → create_prompt()
            → call_llama_api.py → LLaMA 3.1 endpoint
            → Persist & display results (JSON + UI tabs)
```

7. Engineering Details

7.1 PDF Text & Image Extraction (`pdf_extractor.py`)

- Uses Docling for text; cleans common OCR artefacts.
- **Branching logic:** if Docling returns **>200 words** (hyperparameter) we skip image captioning to save compute.
- Otherwise, extracts up to **5 largest images** via PikePDF, resizes to 224×224, captions with BLIP, and base64encodes.
- Outputs `<stem>_dual.json` containing metadata and ordered elements.

CLI:

```
pip install -r requirements.txt
python pdf_extractor.py -i <input.pdf> -o <output_dir>
```

7.2 LLaMA API Wrapper (`call_llama_api.py`)

- Sends prompts via HTTPPOST to <https://ml-llm-api-gpu.use.eks.mcap.sip.dev.cloud.synchronoss.net/api/chat>
- Model: `llama3.1:8b-instruct-q8_0`
- Default temperature=0.2, seed=42 for reproducibility.
- Handles API key securely.

7.3 Orchestration & Prompting (`pdf_pipeline.py`)

- Accepts a single PDF or directory.
- Runs extraction, builds prompts (1000 words), and calls LLaMA.
- Saves `<stem>_prompt_response.json` (single document) and aggregates to `llama_responses.json`.
- **Prompt template** (truncated for brevity):

You are an expert document analyst.

TASKS

1. Summarise the document in 2-3 crisp sentences.
2. Classify into ONE category out of:
advertisement, budget, email, ID, form, handwritten, invoice,
letter, memo, news_article, presentation, questionnaire,
resume, scientific_publication, scientific_report, specification

Return ONLY:

```
{  
  "summary": "...",  
  "classification_label": "..."  
}
```

DOCUMENT Chunks:

```
....  
...  
....
```

◦ **Response template ():**

```
{  
  "summary": "...",  
  "classification_label": "..."  
}
```

*Full template lives in `create_prompt()`.

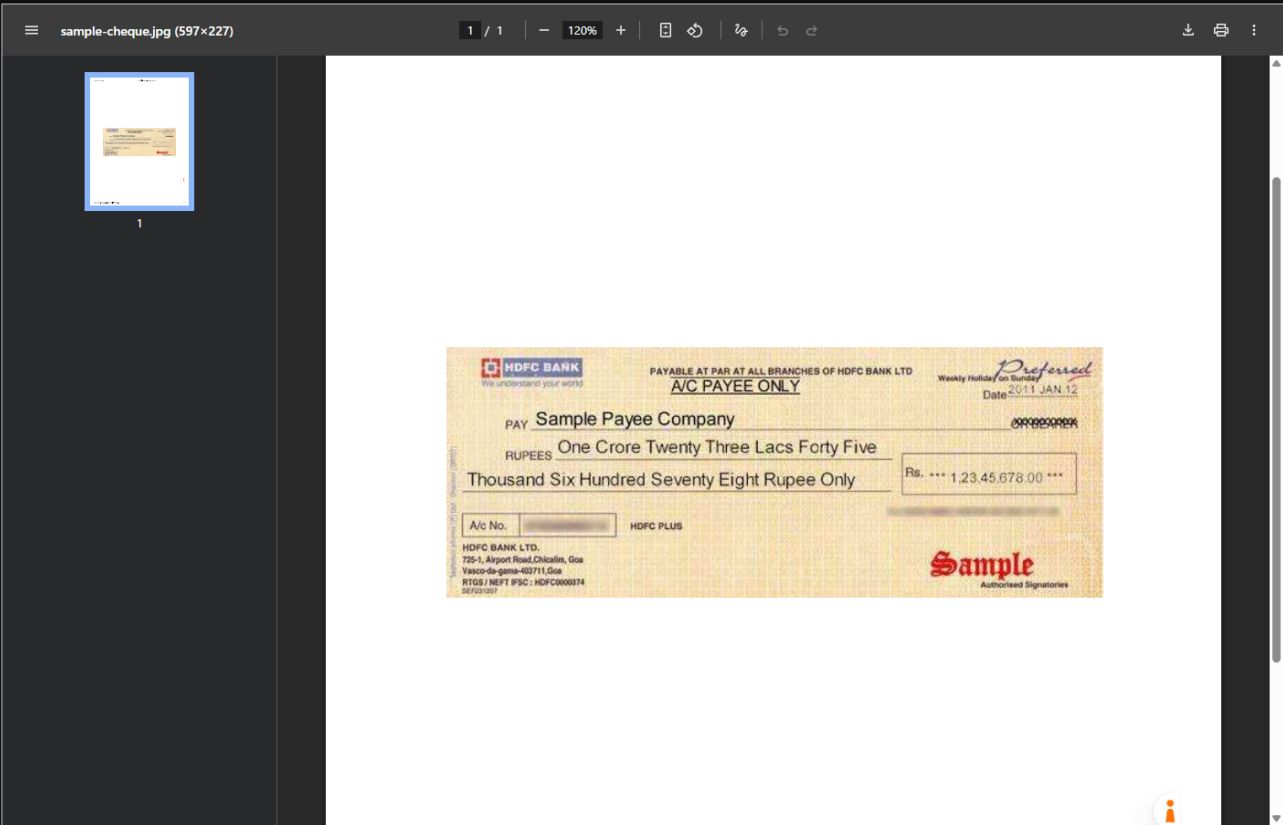
7.4 Interactive UI (`gradio_app.py`)

- Batch upload up to **5 PDFs**.
- Optional result caching (sessionscoped).
- Embedded PDF previews (base64).
- Tabbed views: Summary, Classification, Prompt JSON, Plaintext, Extraction JSON.

8. Results & Evaluation

Processed Documents

C1



LLM Summary

The document appears to be an image-based collection, featuring two images with captions. One image shows a white button with a red button on it, while the other displays a business card with a company name.

Classification Label

advertisement

Processed Documents

DL3

Washington, D.C. sample_driver's license_c_2018

1 / 1 - 102% +

Download Print



1



LLM Summary

A fake ID card with a picture of a woman.



Classification Label

ID

Processed Documents

DL1

ID_driver_GenericTemplate.jpg (1140x750)

1 / 1

131%



1



LLM Summary

This document appears to be a Texas ID card, containing personal information such as height, sex, and eye color. The document includes two images, one of the ID card itself and another of a button with an icon. No other relevant text is present.



Classification Label

ID

Processed Documents

Receipt 1

ZCmVuZHN0cmVhbQplbmRvYmoKNiAwIG9iag08PAovTGluZ3R...

1 / 1

135%

+

🔍

🔄

🔗

📄

📄

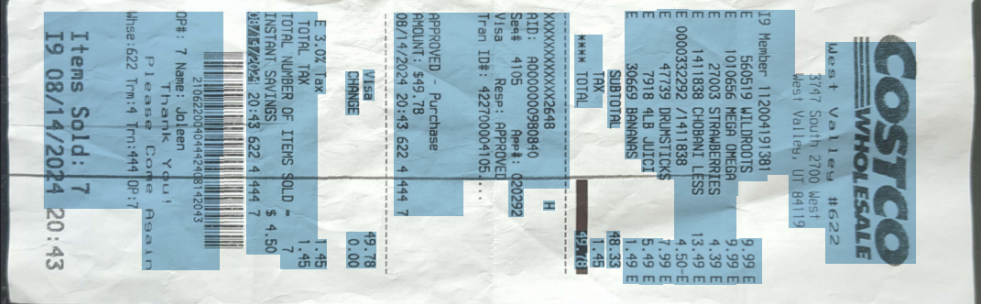
↓

🖨️

⋮



1



LLM Summary

A receipt for a car that is not in service.



Classification Label

invoice

Processed Documents

Business Card 1

9kKZW5kc3RyZWFTcmVuZG9iago2IDAga2JqCjw8Cf9MZW5ndG...

1 / 1

135%



1

Wael H. Mostafa, MD, PhD

Neurosurgeon, Director of Brain Tumor and Skull Base Center

Carle Neuroscience Institute

Carle Physician Group

610 N. Lincoln Avenue | Urbana, IL 61801

(217) 383-6555 | fax (217) 383-7069

carle.org



LLM Summary

Wael H. Mostafa is a neurosurgeon and director of the Brain Tumor and Skull Base Center at Carle Neuroscience Institute.



Classification Label

ID

Processed Documents

Reservation detail 2



Your Chosen Hotel

Vista Hotel ★★★

P. Kakabadze 25, Tbilisi City Centre, Georgia

Customer Support

917349237913

Check-in Time

02:00 PM

Vista Hotel is Located in Set in Tbilisi City, the hotel is featuring family rooms, this property also provides guests with a terrace. It offers 18 guest rooms with garden, mountain, landmark, city views. The hotel provides a continental breakfast at the hotel's bar. Places nearby: Rustaveli Metro Station: 0.6 km Tbilisi Concert Hall: 1 km Freedom Square: 2 km Heroes Square: 2 km Presidential Palace: 2 km Tbilisi Opera and Ballet Theatre: 2 km Rustaveli Theatre: 3 km Tbilisi International Airport: 19 km

Important Information

Your Booking

Reservation Number

177326

Hotel Booking No

107109

Arriving Guests

Mohammad Yeakub

Note

Moodle

Drinn

LLM Summary

This document confirms a hotel reservation for Mohammad Yeakub at Vista Hotel in Tbilisi, Georgia, with check-in on 02-00 PM and a total price of USD 3588.93.



Classification Label

invoice

Processed Documents

R2

ZCmVuZHN0cmVhbQplbmRvYmoKNiAwIG9iago8PAovTG9uZ3R...

1 / 1

135%

🔍 ↺ ⌂ ⏪ ⏩ ⌂

📄 🖨 ⋮



1



LLM Summary

This is a receipt for an online purchase, including tracking numbers and purchase details.



Classification Label

invoice

Processed Documents

Ticket 2

6A2+xlMSINCmVuZHN0cmVhbQplbmRvYmoKOSAwIG9iag08PA...

1 / 1

100%

1

BLACK COFFEE

ELECTRONIC MUSIC

NAME

BLACK COFFEE

DATE

NOV 22, 2022

DOORS OPEN

9:00 PM

SEC ROW SEAT

20 X 88

MESSAGE

ENJOY

ADMIT ONE

LLM Summary

Classification Label

A ticket for the Black Coffee concert on November 22, 2022, at 9:00 PM in section row SEAT 20 X 88

ticket

Prompt & Response (JSON)

Prompt & Response (Text)

Extraction JSON

9. Roadmap / ToDo

1. Extend support to DOC/DOCX and image formats.
2. Expand classification categories beyond the current list.

10. Internal Commands to turn off/on corn Job:

"k8s-sip-eks-mcap-use-dev-default-conf" - this config file is needed - if don't have contact ML engineering team.

Check Service Status:

- **Purpose:** Quick status check to see if service is running, starting, or stopped

```
#Check if pods are running:
./kube/kubectl.exe --kubeconfig="./kube/k8s-sip-eks-mcap-use-dev-default-conf 2" get pods -n ml-ollama

#Check deployment status:
./kube/kubectl.exe --kubeconfig="./kube/k8s-sip-eks-mcap-use-dev-default-conf 2" get deployments -n ml-ollama

#Check cron jobs schedule:
./ kube/kubectl.exe --kubeconfig="./kube/k8s-sip-eks-mcap-use-dev-default-conf 2" get cronjobs -n ml-ollama
```

Start the Service (Scale Up):

- **Purpose:** Manually start the Llama API service when it's down

```
# Scale up main service:
./kube/kubectl.exe --kubeconfig="./kube/k8s-sip-eks-mcap-use-dev-default-conf 2" scale deployment ml-ollama-ollama-gpu-cronjobs --replicas=1 -n ml-ollama

#Scale up auth service
./kube/kubectl.exe --kubeconfig="./kube/k8s-sip-eks-mcap-use-dev-default-conf 2" scale deployment ml-ollama-ollama-gpu-cronjobs-auth --replicas=1 -n ml-ollama
```